



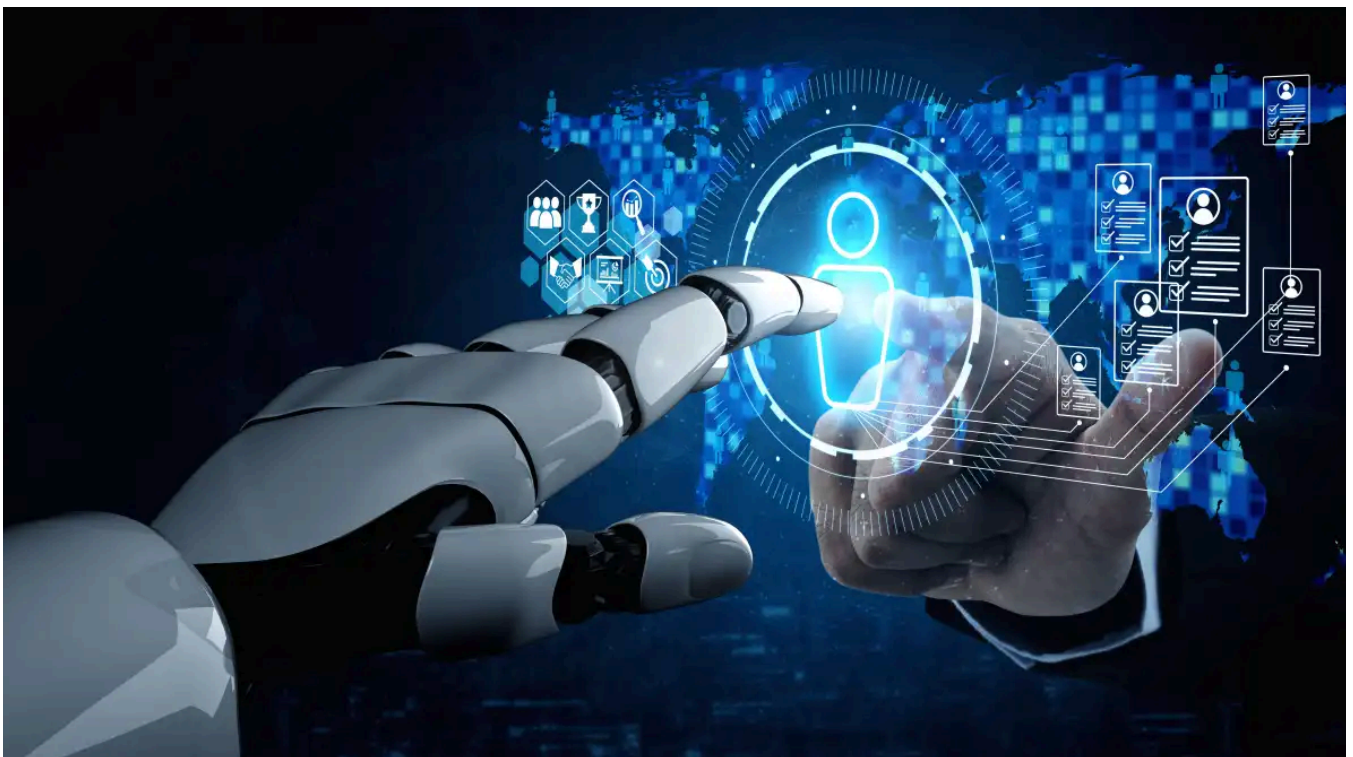
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Microsoft Web IQ: Ground your AI agents with up-to-date web data

Analysis

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Agents need fast, fresh, real-world data to stay grounded and avoid errors. Microsoft's new grounding service helps solve this problem.



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Microsoft has unveiled **a suite of IQ products** [<https://www.microsoft.com/en-us/ai/microsoft-iq#Products>] over the last few months. Work IQ, Fabric IQ, and Foundry IQ build on what Microsoft used to call its “Graphs,” the underlying data architecture that underpins its cloud services. These graphs provided a way to query the data your business uses, treating that data as nodes in a graph database and using the **GraphQL API model** [<https://www.infoworld.com/article/2267992/what-is-graphql-better-apis-by-design.html>] to extract that data — for example, to pull data related to a specific individual held across the various Microsoft 365 applications.

The IQ suite follows a similar approach, using the same data, but treating it as the sparse vector store needed to provide grounding data for **LLM** [<https://www.infoworld.com/article/2335213/large-language-models-the-foundations-of-generative-ai.html>]-based applications. By treating the data as a set of **embedding vectors** [<https://www.infoworld.com/article/2335281/vector-databases-in-llms-and-search.html>], and integrating it with **Model Context Protocol** [<https://www.infoworld.com/article/4029634/what-is-model-context-protocol-how-mcp-bridges-ai-and-external-services.html>] (MCP) servers, Microsoft is giving you the necessary tools to tie LLM output to your data, reducing the risk of hallucination and improving accuracy. Using your own data is a key part of delivering effective agents, ensuring they work within your constraints.

Extending IQ to the web

Web IQ [<https://www.microsoft.com/en-us/webiq>], the latest member of the IQ suite, was unveiled at Build 2026. A modernization of the retired Bing Search APIs, **Web IQ is an agent-focused web search tool** [<https://webiq.microsoft.ai/documentation/overview/>] that builds on the massive Bing search index to provide up-to-date general information for use in your applications.

It may seem a little odd to be talking about a web-wide source of grounding data in the context of a suite of tools that exist to improve the accuracy of your AI applications by providing access to your Microsoft-hosted data. However, in many cases you want to link your agent not only to your data but also to related information from the wider world. For example, an agent powering an ecommerce service could use Web IQ and web-based data sources to provide product comparisons. An agent managing stock levels for a product that is weather-sensitive could use Web IQ as a source of weather data, using Bing’s multiple weather feeds and forecasts.

Just as Google Gemini drew on Google Search, Microsoft Copilot began by using Bing search data to provide grounding for consumer chatbots. It’s easy to take a service like Bing and use it with a LLM, as the nearest neighbor search algorithms use semantic vector similarity techniques to find results that look like your query, ranking them according to their proximity to your search terms.

Microsoft has been tuning its search vector index and the underlying technology stack to work with agents, as agents operate much differently than humans searching the web or querying a chatbot. Providing web search capabilities to agents means having to deal

with persistent queries, as the agent hunts for the information it needs, refining queries and applying reasoning algorithms to develop the response it needs. LLM inferencing requires quick responses that deliver large amounts of data, working with queries that go far beyond the one-word or two-word requests that are typical of humans.

More than the training weights

Using Web IQ gives you access to up-to-date information, beyond the training data used to build and weight an LLM. Bing's crawler works within the standards developed by the search engine industry, obeying meta tags and using its own algorithms to crawl regularly updated websites more often. Bing's crawler ensures that data is both fresh and being used appropriately, with a focus on quality rather than quantity.

Providing access to web data is only part of Web IQ. Microsoft is using Web IQ to host its own models to manage embeddings, ranking, and content extraction, all running on the company's global hyperscale platform. The intent here is to use only a limited number of models, to keep the system performance high while aiming to deliver accurate results. The Web IQ models are different from those used to deliver search results to humans, as they're designed to deliver responses that are suitable for LLMs to use for reasoning.

The underlying search system is based on the DiskANN algorithm [<https://www.microsoft.com/en-us/research/project/project-akupara-approximate-nearest-neighbor-search-for-large-scale-semantic-search/>] developed by Microsoft Research, which allows fast search without requiring enormous amounts of in-memory data access. This approach has been extended to manage information retrieval at scale, building on Microsoft's distributed systems architectures, to support the demands Microsoft is seeing from agent-based systems. At the same time, it must respond to the rapidly changing economics of inference, where token costs now demand the best possible output from the fewest tokens. To meet those economic demands, the Web IQ platform doesn't deliver whole documents to querying agents. Whole documents can lead to expensive inference further down the chain, as LLMs process results repeatedly to drive the agent workflow. Instead, Web IQ structures the information retrieved from the underlying search engine data, delivering what Microsoft calls "structured evidence objects" as well as passage-level information from unstructured text documents. This should result in a much higher signal-to-noise ratio than simply querying a search engine, with a focus on delivering information that lets agents work using fewer tokens.

Using Web IQ in your agent code

The API for Web IQ [<https://webiq.microsoft.ai/documentation/api-reference/web/>] is a standard REST call [<https://webiq.microsoft.ai/documentation/api-reference/web/>], delivering a request object to the Web IQ endpoint. Along with your API authorization key, you will send a query, a set of parameters that control the number of results returned, the language and region used, and the maximum size of the responses and the format used. Responses can

be returned in text, HTML, or markdown formats, as well as extracted passages that are selected for context. All other options return the full document, so can be more expensive to use. Markdown is an interesting alternative, as it can be used as the basis for giving agents semantic memories.

Results include important contextual and citation information, including web page titles and URLs, as well as data about when the site was last crawled and how stale the underlying information is. This can be used to improve grounding and provide more information that can be included in formatted responses — much in the same way as Bing’s Copilot displays context in the form of footnotes in its responses.

Responses to **video searches** [<https://webiq.microsoft.ai/documentation/api-reference/videos/>] include text descriptions. If these aren’t provided as part of the original web content, they will be generated by an LLM. The same approach is used for **image searches** [<https://webiq.microsoft.ai/documentation/api-reference/images/>], with both offering the same contextual cues as the web search API. If you don’t care about the type of data being returned, you can choose a “**classic search** [<https://webiq.microsoft.ai/documentation/api-reference/classic/>],” which will return text, images, video, and news.

Supporting autonomous agents

Microsoft provides LLM-ready documentation for the Web IQ service, with an [llms.txt](#) file and an OpenAPI description. These allow AI tools to discover Web IQ capabilities and include them in workflows as part of autonomous operations, so that agents and other AI applications can implement grounding calls to Web IQ whenever user interactions require them. The API **descriptions include errors** [<https://webiq.microsoft.ai/documentation/error-handling/>] as well as the structure of a standard 200 response.

As Web IQ is designed for use by modern agent frameworks, the Web IQ API is available through an MCP server. The **Web IQ MCP server** [<https://webiq.microsoft.ai/documentation/mcp/>] exposes tools that map to API calls: web, videos, news, and images. They also include a browse option, which lets you pull content from a target URL. The service can be configured with a standard JSON file and requires an API key to control access and manage billing. If your account doesn’t have access to a specific tool, then it won’t be available from inside the MCP server.

If you’re building an agent and you want to evaluate the Web IQ MCP server, it can be added to common coding agents, such as the GitHub Copilot CLI. You can then test it out using familiar tools and generate code that can be dropped into applications via your choice of development tooling. Queries sent to the Web IQ MCP server use the same syntax as REST calls, without having to construct the calls yourself. Working with the MCP server allows you to connect Web IQ to your choice of agent framework, relying on its built-in MCP methods to reduce the code and maintenance overhead.

Web IQ is not for human interactions; Microsoft provides an alternative “**Grounding with Bing** [<https://learn.microsoft.com/en-us/azure/foundry-classic/agents/how-to/tools-classic/bing-grounding?view=azure-python-preview&tabs=python&pivots=overview>]” service for chatbots. Instead, Web IQ is a tool for agents, providing necessary background information that helps keep results fresh and relevant. It’s easy to use, fast, and, above all, cheap, which makes it an ideal tool for modern inference platforms built around Microsoft Azure’s AI tooling.

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